```
#include <iostream>
using namespace std;
const int N = 8;
int board[N][N];
void printBoard() {
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++)
            cout << board[i][j] << " ";
        cout << endl;
    }
}
bool isSafe(int row, int col) {
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            if (board[i][j] == 1) {
                if (j == col) return false;
                if (i - j == row - col) return false;
                if (i + j == row + col) return false;
            }
        }
    }
    return true;
}
bool solve(int row, int firstRow) {
    if (row == N) return true;
    if (row == firstRow)
        return solve(row + 1, firstRow);
    for (int col = 0; col < N; col++) {
        if (isSafe(row, col)) {
            board[row][col] = 1;
            if (solve(row + 1, firstRow)) return true;
            board[row][col] = 0;
        }
    }
    return false;
}
int main() {
    int r, c;
    do {
        cout << "Enter row and column of first queen (0 to 7): ";
        cin >> r >> c;
    } while (r < 0 || r > 7 || c < 0 || c > 7);
    for (int i = 0; i < N; i++)
        for (int j = 0; j < N; j++)
            board[i][j] = 0;
    board[r][c] = 1;
    if (solve(0, r)) {
        int result[N][N];
        for (int i = 0; i < N; i++)
            for (int j = 0; j < N; j++) {
                result[i][j] = board[i][j];
                board[i][j] = 0;
            }
        int step = 1;
        board[r][c] = 1;
        cout << "\nStep 1: First queen placed at row " << r << ", column " << c << endl;
        printBoard();
        for (int i = 0; i < N; i++) {
            if (i == r) continue;
            for (int j = 0; j < N; j++) {
                if (result[i][j] == 1) {
                    board[i][j] = 1;
                    step++;
                    cout << "\nStep " << step << ": Queen placed at row " << i << ", column " << j << endl;
                    printBoard();
                }
            }
        }
        cout << "\nAll 8 queens are placed.\n";
    } else {
        cout << "No solution possible with the first queen at this position.\n";
    }
    return 0;
}
```

```
PS C:\Users\fishb\Desktop\Sem 7\DAA> cd "c:\Users\fishb\Desktop\Sem 7\DAA\" ; if ($?) { g++ 6_eight_queens.cpp -o 6_eight
_queens } ; if ($?) { .\6_eight_queens }
Enter row and column of first queen (0 to 7): 0 2

Step 1: First queen placed at row 0, column 2
0 0 1 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0

Step 2: Queen placed at row 1, column 0
0 0 1 0 0 0 0 0
1 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0

Step 3: Queen placed at row 2, column 6
0 0 1 0 0 0 0 0
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0

PS C:\Users\fishb\Desktop\Sem 7\DAA> cd "c:\Users\fishb\Desktop\Sem 7\DAA\" ; if ($?) { g++ 6_eight_queens.cpp -o 6_eight
_queens } ; if ($?) { .\6_eight_queens }
Step 4: Queen placed at row 3, column 4
0 0 1 0 0 0 0 0
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0

Step 5: Queen placed at row 4, column 7
0 0 1 0 0 0 0 0
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 1
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0

Step 6: Queen placed at row 5, column 1
0 0 1 0 0 0 0 0
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 1
0 1 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0
```

```
PS C:\Users\fishb\Desktop\Sem 7\DAA> cd "c:\Users\fishb\Desktop\Sem 7\DAA\" ; if ($?) { g++ 6_eight_queens.cpp -o 6_eight
_queens } ; if ($?) { .\6_eight_queens }
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 1
0 1 0 0 0 0 0 0
0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0

Step 7: Queen placed at row 6, column 3
0 0 1 0 0 0 0 0
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 1
0 1 0 0 0 0 0 0
0 0 0 1 0 0 0 0
0 0 0 0 0 0 0 0

Step 8: Queen placed at row 7, column 5
0 0 1 0 0 0 0 0
1 0 0 0 0 0 0 0
0 0 0 0 0 0 1 0
0 0 0 0 1 0 0 0
0 0 0 0 0 0 0 1
0 1 0 0 0 0 0 0
0 0 0 1 0 0 0 0
0 0 0 0 0 1 0 0

All 8 queens are placed.
```